1. **1 point.** (RL2e 3.25 – 3.29) *Fun with Bellman.*
Written: Write the Bellman equations for the value functions in terms of the three-argument function $p(s'|s, a)$ (Equation 3.4) and the two-argument function $r(s, a)$ (Equation 3.5).

   (a) Give an equation for v_* in terms of q_*.

   (b) Give an equation for q_* in terms of v_* and p.

   (c) Give an equation for π_* in terms of q_*.

   (d) Give an equation for π_* in terms of v_* and p.

$$V^*(s) = \mathbb{E}_{s'}\left[r(s,a) + \gamma V^*(s')\right]$$

a) $V^*(s) = Q^*(s, \pi^*(s))$

b) $Q^*(s,a) = \sum_{s'} p(s'|s,a)\left[r(s,a) + \gamma V^{\pi}(s')\right]$

c) $\pi^*(s) = \underset{a}{\operatorname{argmax}}\ Q^*(s,a)$

d) $\pi^*(s) = \underset{a}{\operatorname{argmax}} \sum_{s'} p(s'|s,a)\left[r(s,a) + \gamma V^{\pi}(s')\right]$

2. **3 points.** *Policy iteration by hand.*
<u>**Written:**</u> Consider an undiscounted MDP having three states, x, y, z. State z is a terminal state. In states x and y there are two possible actions: b and c. The transition model is as follows:

- In state x, action b moves the agent to state y with probability 0.7 and makes the agent stay put (at state x) with probability 0.3.
- In state y, action b moves the agent to state x with probability 0.7 and makes the agent stay put (at state y) with probability 0.3.
- In either state x or state y, action c moves the agent to state z with probability 0.2 and makes the agent stay put with probability 0.8.

The reward model is as follows:

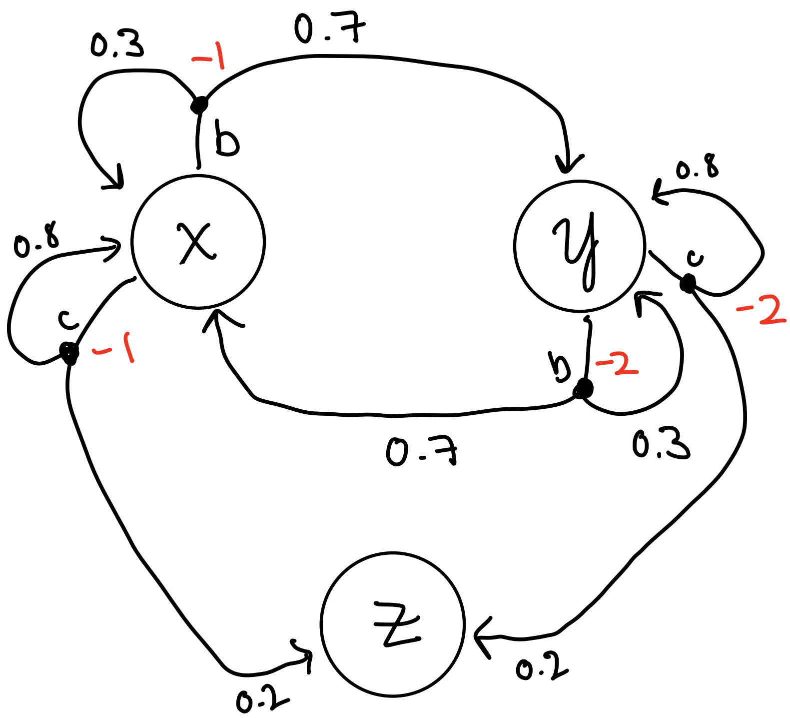
- In state x, the agent receives reward -1 regardless of what action is taken and what the next state is.
- In state y, the agent receives reward -2 regardless of what action is taken and what the next state is.

Answer the following questions:

(a) What can be determined *qualitatively* about the optimal policy in states x and y (i.e., just by looking at the transition and reward structure, *without* running value/policy iteration to solve the MDP)?

(b) Apply policy iteration, showing each step in full, to determine the optimal policy and the values of states x and y. Assume that the initial policy has action c in both states.

(c) What happens to policy iteration if the initial policy has action b in both states? Does a discounting factor (for example $\gamma = 0.9$) help? Does the optimal policy change with different γ in this particular MDP?



a) Qualitatively,

if the agent starts in state y,
the most optimal policy will be to take action b
first make sure the agent gets to state x.

once in state x, the most optimal policy is to take
action c, to eventually get to the terminal state.

Policy Eval #1

$v^0(x) = 0 \quad v(y) = 0$

$v^0(x) = p(x|x,c)\left[-1 + v(x)\right] + p(z|x,c)\left[-1 + v(z)\right]$

$= 0.8[-1] + 0.2[-1]$

$v^1(x) = -1$

$v^1(y) = -2$

$v^2(x) = 0.8[-1-1] + 0.2[-1]$

$= -1.8$

$v^3(x) = -2.44$

$v4(x) = -2.95$

$v^5(x) = -3.36$

$v^6(x) = -3.69$

$v^7(x) = -3.95$

$= -4.16$

$= -4.33$

$= -4.46$

$= -4.57$

$\vdots$

$v^*(x) = -5$

$v^2(y) = 0.8[-2-2] + 0.2[-2]$

$= -3.6$

$v^3(y) = -4.88$

$v^4(y) = -5.91$

$v^5(y) = -6.72$

$v^6(y) = -7.38$

$\vdots$

$v^*(y) = -10$

Initial Results Can also be found Algebraicly

$V^\pi(x) = 0.2\left[-1 + V^\pi(z)\right] + 0.8\left[-1 + V^\pi(x)\right]$

$V^\pi(x) = -0.2 - 0.8 + 0.8V^\pi(x)$

$V^\pi(x) = -1 + 0.8V^\pi(x)$

$V^\pi(x) = \frac{-1}{0.2} = -5$

$V^\pi(y) = 0.2\left[-2 + V^\pi(z)\right] + 0.8\left[-2 + V^\pi(y)\right]$

$V^\pi(y) = -0.4 - 1.6 + 0.8V^\pi(y)$

$V^\pi(y) = -2 + 0.8V^\pi(y)$

$V^\pi(y) = \frac{-2}{0.2} = -10$

Policy Iteration #1:

$$\pi(x) = \arg\max_a Q^\pi(x,a)$$

$$Q^\pi(x,b) = \sum_{s'} p(s'|s,a)\left[r(s,a) + \gamma V^\pi(s')\right]$$

$$= p(x|x,b)\cdot[-1 + V^\pi(x)]$$

$$+$$

$$p(y|x,b)\cdot[-1 + V^\pi(y)]$$

$$= 0.3[-1-5] + 0.7[-1-10]$$

$$= -1.8 - 7.7$$

$$Q^\pi(x,b) = -9.5$$

$$^\pi(x,c) = p(x|x,c)\cdot[-1 + V^\pi(x)]$$

$$+ \; p(z|x,c)\cdot[-1 + \cancel{V^\pi(z)}]$$

$$= 0.8[-1-5] + 0.2[-1]$$

$$= -0.8 - 4 - 0.2$$

$$Q^\pi(x,c) = -5$$

$\pi^*(x) = \arg\max_a Q^\pi(x,a) =$ <u>action c</u>

$$\pi^*(y) = \arg\max_a Q^\pi(y,a)$$

$$Q^\pi(y,b) = p(y|y,b)\cdot[-2 + V^\pi(y)]$$

$$+$$

$$p(x|y,b)\cdot[-2 + V^\pi(x)]$$

$$= 0.3[-2-10] + 0.7[-2-5]$$

$$= -3.6 - 4.9$$

$$Q^\pi(y,b) = -8.5$$

$$Q^\pi(y,c) = p(y|y,c)\cdot[-2 + V^\pi(y)]$$

$$+$$

$$= p(z|y,c)[-2 + \cancel{V^\pi(z)}]$$

$$= 0.8[-2-10] + 0.2[-2]$$

$$= -9.6 - 0.4$$

$$Q^\pi(y,c) = -10$$

$\pi^*(y) = \arg\max_a Q^\pi(y,a) =$ Action b

Policy Eval #2

$\pi(x) = c \quad \pi(y) = b$

$v^0(x) = -5 \quad v^0(y) = -10$

$v^1(x) = p(x|x,c)[-1 + V(x)] + p(z|x,c)[-1 + V(z)]$

$= 0.8[-1-5] + 0.2[-1]$

$= -5$ → Fully Converged (doesn't depend on y)

$v^1(y) = p(y|y,b)[-2 + V(y)] + p(x|y,b)[-2 + V(x)]$

$= 0.3[-2-10] + 0.7[-2-5]$

$= -3.6 - 4.9$

$= -8.5$

$v^2(y) = 0.3[-2-8.5] - 4.9$

$= -8.05$

$v^3(y) = -7.915$

$v^4(y) = -7.87$

$v^5(y) = -7.86$

$v^6(y) = -7.86$ ✓

Policy Iteration #2

$$\pi(x) = \underset{a}{\operatorname{argmax}}\, Q^{\pi}(x,a)$$

$$Q^{\pi}(x,b) = \sum_{s'} p(s'|s,a)\left[r(s,a) + \gamma V^{\pi}(s')\right]$$

$$= p(x|x,b)\cdot\left[-1 + V^{\pi}(x)\right] + p(y|x,b)\cdot\left[-1 + V^{\pi}(y)\right]$$

$$= 0.3[-1-5] + 0.7[-1-10]$$

$$= -1.8 - 7.7$$

$$Q^{\pi}(x,b) = -9.5$$

$$Q^{\pi}(x,c) = p(x|x,c)\cdot\left[-1 + V^{\pi}(x)\right] + p(z|x,c)\cdot\left[-1 + \cancel{V^{\pi}(z)}\right]$$

$$= 0.8[-1-5] + 0.2[-1]$$

$$= -0.8 - 4 - 0.2$$

☆ $Q^{\pi}(x,c) = -5$

$$\pi^{*}(x) = \underset{a}{\operatorname{argmax}}\, Q^{\pi}(x,a) = \underline{\text{action } c}$$

$$\pi(y) = \underset{a}{\operatorname{argmax}}\, Q^{\pi}(y,a)$$

$$Q^{\pi}(y,b) = p(y|y,b)\cdot\left[-2 + V^{\pi}(y)\right] + p(x|y,b)\cdot\left[-2 + V^{\pi}(x)\right]$$

$$= 0.3[-2-7.86] + 0.7[-2-5]$$

$$= -2.96 - 4.9$$

☆ $Q^{\pi}(y,b) = -7.86$

$$Q^{\pi}(y,c) = p(y|y,c)\cdot\left[-2 + V^{\pi}(y)\right] + p(z|y,c)\left[-2 + \cancel{V^{\pi}(z)}\right]$$

$$= 0.8[-2-7.86] + 0.2[-2]$$

$$Q^{\pi}(y,c) = -8.29$$

$$\pi^{*}(y) = \underset{a}{\operatorname{argmax}}\, Q^{\pi}(y,a) = \text{Action } b$$

Since there was no change to the policy, policy iteration has converged to the optimal policy.

c) If the initial action was b for both states, and $\gamma = 1$, the policy evaluation will never converge without a terminal state, the values keep approaching $-\infty$

$V(x) = 0 \quad V(y) = 0$

$V(x) = p(x|x,b)[-1 + V(x)] + p(y|x,b)[-1 + V(y)]$

$= 0.3[-1] + 0.7[-1]$

$V(x) = -1 \quad V(y) = -2$

$V(x) = 0.3[-1-1] + 0.7[-1-2] = -0.6 - 2.1 = -2.7$

$V(y) = 0.3[-2-2] + 0.7[-2-1] = -1.2 - 2.1 = -3.3$

$\vdots$

However, if $0 < \gamma < 1$, this discounting means that the value function will not approach infinity but will instead taper off eventually.

3.

a) Value Iteration Output

```
Value Iteration converged after 43 iterations.
=============================
==  Optimal State Value    ==
=============================
[[11.9 14.9 11.9 10.2  8.2]
 [ 9.5 11.9  9.5  8.2  6.6]
 [ 7.6  9.5  7.6  6.6  5.2]
 [ 6.1  7.6  6.1  5.2  4.2]
 [ 4.9  6.1  4.9  4.2  3.4]]
=============================
=============================
==     Optimal Policy      ==
=============================
[0, 0] = ['east']
[0, 1] = ['north', 'south', 'west', 'east']
[0, 2] = ['west']
[0, 3] = ['north', 'south', 'west', 'east']
[0, 4] = ['west']
------------------------------
[1, 0] = ['north', 'east']
[1, 1] = ['north']
[1, 2] = ['north', 'west']
[1, 3] = ['north']
[1, 4] = ['north', 'west']
------------------------------
[2, 0] = ['north', 'east']
[2, 1] = ['north']
[2, 2] = ['north', 'west']
[2, 3] = ['north']
[2, 4] = ['north', 'west']
------------------------------
[3, 0] = ['north', 'east']
[3, 1] = ['north']
[3, 2] = ['north', 'west']
[3, 3] = ['north']
[3, 4] = ['north', 'west']
------------------------------
[4, 0] = ['north', 'east']
[4, 1] = ['north']
[4, 2] = ['north', 'west']
[4, 3] = ['north']
[4, 4] = ['north', 'west']
------------------------------
```

b) Policy Iteration Output

```
=============================
==  Optimal State Value    ==
=============================
[[11.9 14.9 11.9 10.2  8.2]
 [ 9.5 11.9  9.5  8.2  6.6]
 [ 7.6  9.5  7.6  6.6  5.2]
 [ 6.1  7.6  6.1  5.2  4.2]
 [ 4.9  6.1  4.9  4.2  3.4]]
=============================
=============================
==     Optimal Policy      ==
=============================
[0, 0] = ['east']
[0, 1] = ['north', 'south', 'west', 'east']
[0, 2] = ['west']
[0, 3] = ['north', 'south', 'west', 'east']
[0, 4] = ['west']
------------------------------
[1, 0] = ['north', 'east']
[1, 1] = ['north']
[1, 2] = ['north', 'west']
[1, 3] = ['north']
[1, 4] = ['north', 'west']
------------------------------
[2, 0] = ['north', 'east']
[2, 1] = ['north']
[2, 2] = ['north', 'west']
[2, 3] = ['north']
[2, 4] = ['north', 'west']
------------------------------
[3, 0] = ['north', 'east']
[3, 1] = ['north']
[3, 2] = ['north', 'west']
[3, 3] = ['north']
[3, 4] = ['north', 'west']
------------------------------
[4, 0] = ['north', 'east']
[4, 1] = ['north']
[4, 2] = ['north', 'west']
[4, 3] = ['north']
[4, 4] = ['north', 'west']
------------------------------
```

4. **1 point.** (RL2e 4.4) *Fixing policy iteration.*
Written:

(a) The policy iteration algorithm on page 80 has a subtle bug in that it may never terminate if the policy continually switches between two or more policies that are equally good. This is okay for pedagogy, but not for actual use. Modify the pseudocode so that convergence is guaranteed.

(b) Is there an analogous bug in value iteration? If so, provide a fix; otherwise, explain why such a bug does not exist.

a) This issue can be fixed by adding a few extra steps to policy improvement

Policy Iteration (using iterative policy evaluation) for estimating $\pi \approx \pi_*$

1. Initialization
   $V(s) \in \mathbb{R}$ and $\pi(s) \in \mathcal{A}(s)$ arbitrarily for all $s \in \mathcal{S}$; $V(terminal) \doteq 0$

2. Policy Evaluation
   Loop:
      $\Delta \leftarrow 0$
      Loop for each $s \in \mathcal{S}$:
         $v \leftarrow V(s)$
         $V(s) \leftarrow \sum_{s',r} p(s',r|s,\pi(s))[r + \gamma V(s')]$
         $\Delta \leftarrow \max(\Delta, |v - V(s)|)$
   until $\Delta < \theta$ (a small positive number determining the accuracy of estimation)

3. Policy Improvement
   policy-stable $\leftarrow$ *true*
   For each $s \in \mathcal{S}$:
      old-action $\leftarrow \pi(s)$

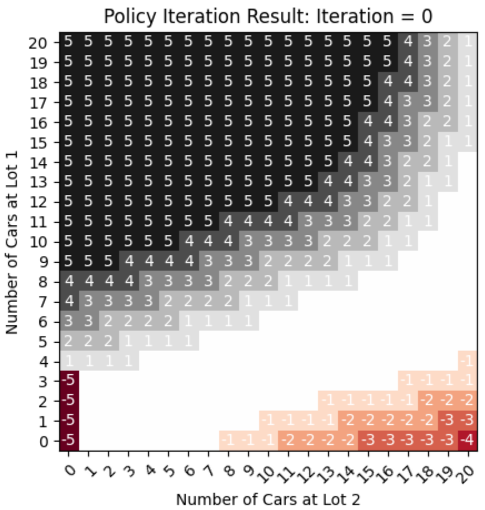
old_q_val = Q(s, old_action)
new_action ← $\underset{a}{\operatorname{argmax}}$ Q(s,a)
new_q_val = Q(s, new_action)
if |old_q_val − new_q_val| > ε :
  π(s) = new_action
else:
  π(s) = old_action

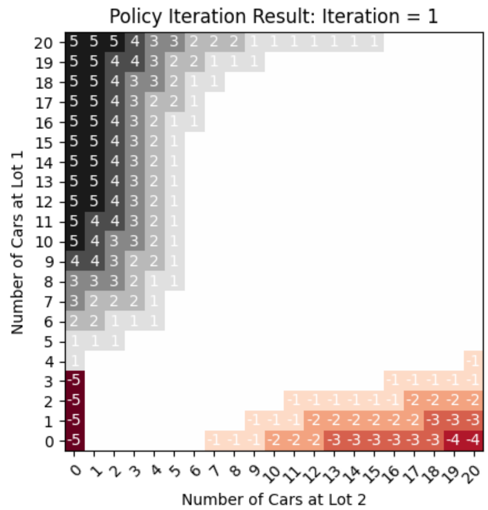
epsilon is some small threshold value

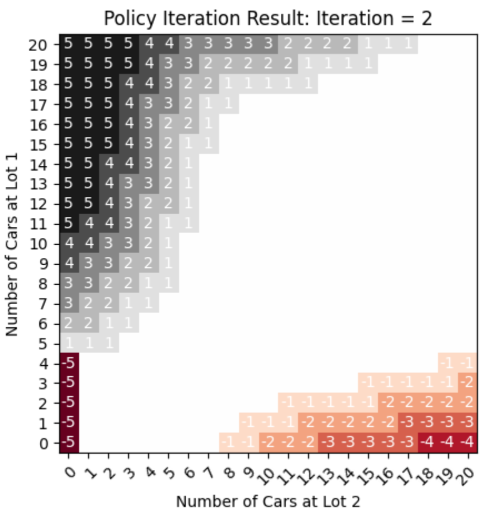
      If *old-action* $\neq \pi(s)$, then *policy-stable* $\leftarrow$ *false*
   If *policy-stable*, then stop and return $V \approx v_*$ and $\pi \approx \pi_*$; else go to 2

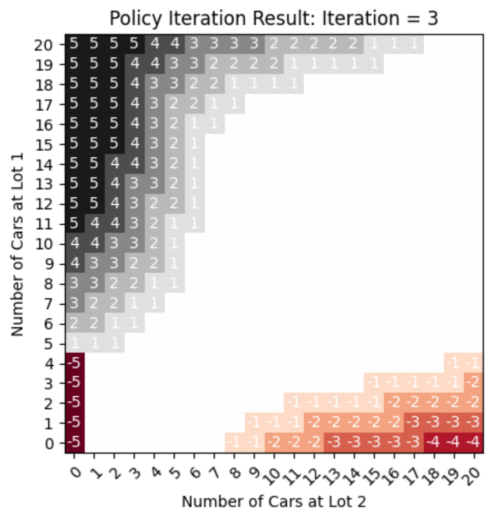
b) This same error could not happen for value iteration because the policy is only generated at the end, after the value function has been finalized, so there will always only be 1 action that satisfies $\underset{a}{\operatorname{argmax}}$ Q(s,a)

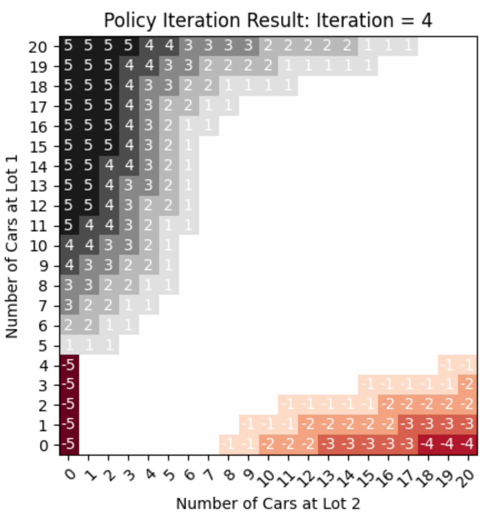
5. a) Rental Car Problem (unmodified reward)

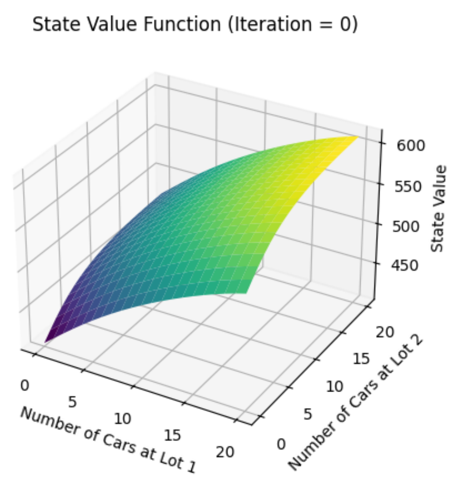




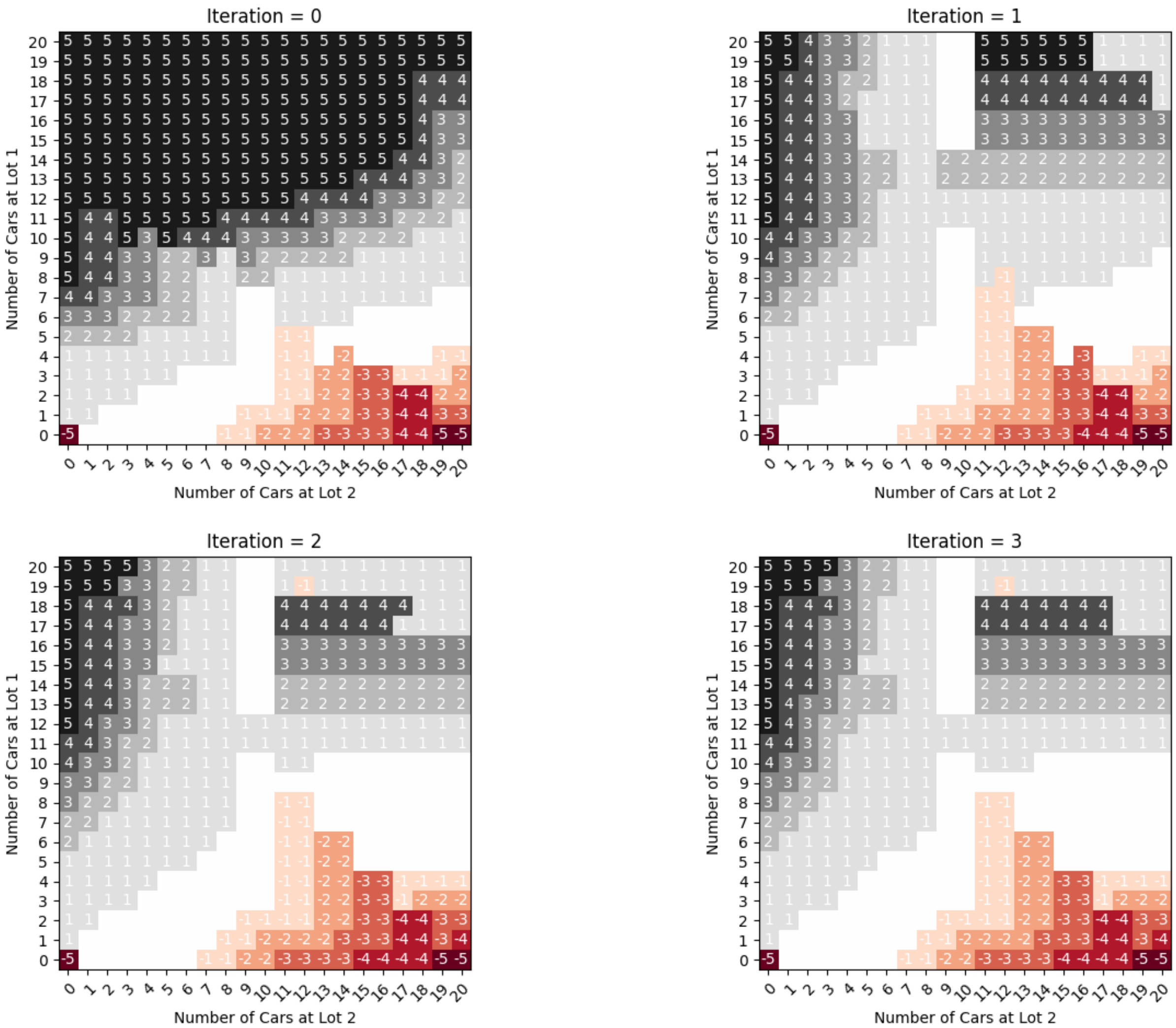


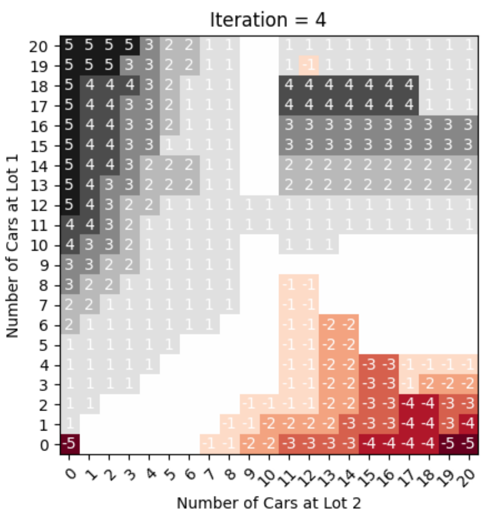


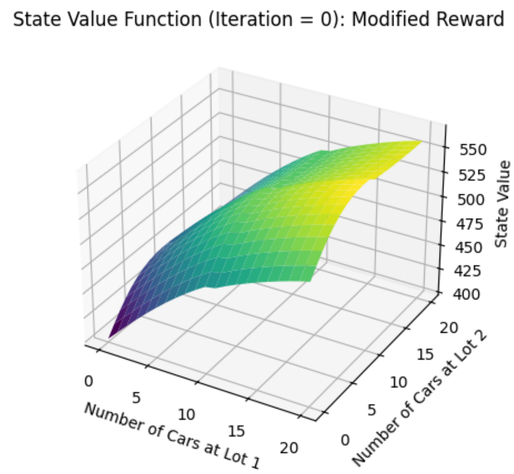




b) Rental Car Problem (Modified Reward)







With the unmodified reward if both lots had a lot of cars, but not the maximum amount, there was no incentive to move them.

With the modified reward, avoiding the overflow parking penalty means that cars get moved from lot 1 to lot 2 to ensure the overflow penalty only occurs for 1 locations. And it moves from lot 1 to lot 2 and not vice versa because of the one free move produced by the modified reward function